KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026-2027 (ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST – I
(Regulations 2022)

Month and Year	July 2026	Roll Number	27TR133.
Programme	B.E./BTech	Date	08.07.2026
Branch	Common	Time	09.15am – 10.45am
Semester	VII	Duration	1½ Hours
Course Code	22MAO09	Max. Marks	50
Course Name	Optimization for Engineers		

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

1. Explain the mathematical formulation of a linear programming problem. (2) CO1 K2
2. Egg contains 6 units of vitamin A per gram and 7 units of vitamin B per gram and cost Rs.12 per gram. Milk contains 8 units of vitamin A per gram and 12 units of vitamin B per gram and cost Rs.20 per gram. The daily minimum requirement of vitamin A and vitamin B are 100 units and 120 units respectively. Formulate this as a LPP (2) CO1 K2
3. Infer Slack Varialble in Linear Programming Problem. (2) CO1 K2
4. Solve the system by Graphical method: (2) CO1 K3
   $Maximize\ Z = 10x_1 + 12x_2$
   $subject\ to\ the\ constraints\ x_1 - x_2 \leqslant -5;\ -x_1 + x_2 \leqslant 0$
   $and\ x_1, x_2 \geqslant 0$
5. Infer feasible solution in Linear programming problem (2) CO1 K2
6. Translate the following system into standard LPP (2) CO1 K2
   $Minimize\ Z = 8x + 9y$
   $subject\ to\ 2x + 3y \leqslant 10;\ 5x + 6y \geqslant 12$
   $and\ x, y \geqslant 0$
7. Solve by graphical method: (2) CO1 K2
   $Minimize Z = 4x_1 + 6x_2\ subject\ to\ the\ constraints\ x_1 - x_2 \leqslant 4;\ x_1 + x_2 \geqslant 6$
   $and\ x_1, x_2 \geqslant 0$
8. Illustrate the types of Integer programming problem and explain. (2) CO2 K2
9. Summerize the importance of Integer programming problem. (2) CO2 K2
10. Extend the LPP into integer programming problem using Gomory cut method. (2) CO2 K2

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. A software development company is planning the allocation of two types of cloud computing services for a large-scale web application. Let x represent the number of Standard Cloud Server instances and y represent the number of High-Performance Cloud Server instances to be deployed. The monthly operating cost of a Standard Server is Rs .20 (lakh), while that of a High-Performance Server is Rs.₹10 (lakh). The company aims to minimize the total monthly infrastructure cost while ensuring that the application meets the required performance and reliability standards. The combined computing capacity, the weighted resource availability and the total network and processing capability is given as (10) CO1 K3
    $Minimize Z = 20x + 10y$
    subject to the constraints
    $x + 2y \geqslant 40;\ 3x + y \geqslant 30;\ 4x + 3y \geqslant 60$
    $and\ x, y \geqslant 0$
    Solve the Linear Programming Problem using the graphical methodto determine the optimal number of Standard and High-Performance Cloud Server instances that minimizes the total monthly operating cost.

12. .Solve the following LPP by simplex method: (10) CO1 K3

$Maximize Z = x_1 + x_2 + 3x_3$

subject to the constraints

$3x_1 + 2x_2 + x_3 \leqslant 3; 2x_1 + x_2 + 2x_3 \leqslant 2;$

$and\ x_1, x_2 \geqslant 0$

13. Solve the following LPP by penalty method: (10) CO1 K3

$Minimize Z = 2x_1 + 3x_2$

subject to the constraints

$x_1 + x_2 \geqslant 5; x_1 + 2x_2 \geqslant 6$

$and\ x_1, x_2 \geqslant 0$

14. A software company is planning the deployment of two types of software modules for a client project. Let x1 denote the number of **Database Management Modules** and x2 denote the number of **Web Service Modules** to be deployed. Each Database Management Module contributes one unit of project value, and each Web Service Module also contributes one unit. The company aims to **maximize the overall project value** while operating within the available development and testing resources. (10) CO2 K3

$Maximize Z = x_1 + x_2$

subject to the constraints

$2x_1 + 5x_2 \leqslant 16; 6x_1 + 5x_2 \leqslant 30$

$and\ x_1, x_2 \geqslant 0$

Solve the integer programming problem with the decision variables must take integer values.

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	30	70	0	0	0

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CONTINUOUS ASSESSMENT TEST - II
Regulation 2022

Month and Year	August 2026	Roll Number	23ITR133.
Programme	BE / BTech	Date	19/08/2026
Branch	II,CSD	Time	9:15 AM - 10:45 AM
Semester	S7	Duration	1 Hour 30 Min
Course Code	22MAO09	Max. Marks	50
Course Name	Optimization for Engineers		

PART A 10 X 2 =20 MARKS
Answer all the questions

1. Infer Mixed Integer Programming Problem. (2) CO2 K2
2. Compare Pure and Mixed Integer Programming Problem. (2) CO2 K2
3. Elabrate the Gomory's fractional cut corresponding to the equation $x_1 + \frac{2}{3}x_3 - \frac{1}{3}x_4 = \frac{2}{3}$ that appears in mixed IPP. (2) CO2 K2
4. Infer Dynamic Programming Problem. (2) CO3 K2
5. Summarize Bellman's Principle of optimality. (2) CO3 K2
6. Outline any two applications of Dynamic Programming Problem. (2) CO3 K2
   Explain when the function y=f(x) attains a minimum. *(2) CO3 K2*
8. Summarize any two characteristics of Dynamic programming Problem. *(2) CO3 K2*
9. Extract first stage recursive equation from (2) CO3 K3
   $Maximize\ b_1x_1 + b_2x_2 + \ldots + b_nx_n$
   $where\ x_1 + x_2 + \ldots + x_n = c.$
10. Differentiate forward computational procedure and backword computational procedure (2) CO3 K2

PART B

PART B

11. Use Branch and Bound method to solve the IPP: (10) CO2 K3
    $max\ Z = 7x_1 + 9x_2$
    $subject\ to - x_1 + 3x_2 \leqslant 6$
    $7x_1 + x_2 \leqslant 35$
    $and\ x_1, x_2\ are\ non - negative\ integers.$
12. Solve the Dynamic Programming Problem (10) CO3 K3
    $Minimize Z = p_1 log p_1 + p_2 log p_2 + \ldots + p_n log p_n\ subject\ to\ p_1 + p_2 + \ldots + p_n = 1\ and\ p_i \geqslant 0$
13. Use Dynamic Programming to solve (10) CO3 K3
    $Maximize\ Z = y_1 . y_2 . y_3$
    $subject\ to\ y_1 + y_2 + y_3 = 5$
    $and\ y_1, y_2, y_3 \geqslant 0$
14. Solve the following LPP by Dynamic Programming approach: (10) CO3 K3
    $Max\ Z = 4x_1 + 14x_2$
    $subject\ to\ 2x_1 + 7x_2 \leq 21$
    $7x_1 + 2x_2 \leqslant 21$
    $and\ x_1, x_2 \geqslant 0$

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	30	70	0	0	0

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CONTINUOUS ASSESSMENT TEST - III
Regulation 2022

Month and Year	October 2026	Roll Number	23IT R123
Programme	BE/BTech	Date	09/10/2026
Branch	IT,CSD	Time	9:15 AM - 10:45 AM
Semester	S7	Duration	1hr30mts
Course Code	22MAO09	Max. Marks	50
Course Name	Optimization for Engineers		

PART-A

Answer all questions.

1. Infer Kendal's Notation in Queing theory. (2) CO4 K2
2. In the usual notation of $(M/M/1):(\infty/FIFO)$ queue system, if $\lambda = 12$per hour and $\mu = 24$ per hour, find the average number of customers in the system. (2) CO4 K2
3. List down the little's formula that hold good for the infinite capacity poisson queue models. (2) CO4 K2
4. In a 3 server infinite capacity poisson queue model if $\frac{\lambda}{\mu s} = \frac{2}{3}$, Find P_0 (2) CO4 K2
5. List the formula used to calculate P_0 in a multi-server, limited-capacity queueing system. (2) CO4 K2
6. Summarize the mathematical formulation of a Non Linear Programming Problem. (2) CO4 K2
7. Show whether the function $f(x) = x^4 + 6x^2 + 12x$ is convex or concave (2) CO5 K2
8. Classify whether the function $g(x) = x_1x_2 - x_1^2 - x_2^2$ is convex, concave or neither (2) CO4 K2
9. Explain the sufficiency of Kuhn-Tucker Condition (2) CO4 K2
10. Extract Q from the following functions:
    $f(x) = x_1^2 - 10x_1 + x_2^2 - 6x_2 + x_3^2 - 4x_3, h(x) = x_1 + x_2 + x_3 = 7$ (2) CO5 K3

PART-B

Answer any 3 questions. Each question carries 10 marks.

11. In a computer science laboratory, a server processes incoming computational requests. Each request takes exactly 10 minutes to be processed, after which the server becomes available to process the next request. Requests arrive according to a Poisson process at an average rate of one request every 27 minutes. Assuming the system has a **single server** and an unlimited waiting queue, calculate the following: (10) CO4 K3
    - The average number of requests in the system.
    - The average number of requests waiting in the queue.
    - The average waiting time of a request in the queue.

12. At a university computer help desk, students arrive according to a Poisson process at an average rate of 8 students per hour. The help desk has two technicians, and each technician can serve an average of 5 students per hour. (10) CO4 K3
    Assuming that the service times are exponentially distributed and that there is unlimited waiting space, calculate:
    - The probability that there are no students in the system.
    - The average number of students waiting in the queue.
    - The average number of students in the system.
    - The average time a student spends in the system.

13. Solve the Non Linear Programming Problem: (10) CO5 K3
    $Minimize\ Z = x_1^2 + x_2^2 + x_3^2$
    $Subject\ 4x_1 + x_2^2 + 2x_3 = 14\ \ and\ x_1, x_2, x_3 \geqslant 0$

14. Solve the NLPP: (10) CO5 K3
    $Maximize\ Z = 10x_1 + 4x_2 - 2x_1^2 - x_2^2$
    $subject\ to\ 2x_1 + x_2 \leqslant 5$
    $x_1, x_2 \geq 0$

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	30	70	0	0	0